

CENTER-ZERO MAGIC SQUARES OF SQUARES OVER NUMBER FIELDS

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ABSTRACT. We study proper 3×3 magic squares whose entries are distinct squares in a number field and whose center entry is 0. For a proper center-zero square, Lucas's parametrization forces the ground field to contain $i = \sqrt{-1}$ and reduces existence to the presence of a point over the ground field outside an explicit finite degeneracy locus on the genus-one curve $C \subset \mathbb{P}_{[u:y:z:w]}^3$ defined by $y^2 = u^2 + w^2$ and $z^2 = u^2 - w^2$. With a rational point chosen as origin, C is an elliptic curve over $\mathbb{Q}$ isogenous to the congruent-number curve $E: y^2 = x^3 - x$. We prove that $C(\mathbb{Q}(i))$ consists of exactly eight points, all degenerate; consequently, no quadratic number field admits a proper center-zero magic square of squares. Conversely, if n is a congruent number, $E_n: y^2 = x^3 - n^2x$, and $K = \mathbb{Q}(i, \sqrt{n})$, then $\text{rank } E(K) = 2 \text{rank } E_n(\mathbb{Q}) \geq 2$. Rational right triangles of area n thus produce infinitely many magic squares of squares over K , even modulo scaling and grid symmetries; hence degree 4 is minimal. For $n = 5$, we give an explicit example arising from the arithmetic progression $31^2, 41^2, 49^2$.

1. INTRODUCTION

The classical 3×3 magic square of squares problem asks whether nine distinct integer squares can be placed in a 3×3 grid so that all three rows, all three columns, and both main diagonals have the same sum. Magic squares of squares themselves go back to Euler, who exhibited a fully magic 4×4 example in a letter to Lagrange of 1770. The 3×3 problem goes back at least to Lucas, who took up Euler's question in 1876 [14]; it was posed again by LaBar [12] and later popularized by Gardner [6]. The problem remains open: see Boyer [1] for a survey of the problem's history and computational record; and Robertson [18] for the standard dictionary relating the problem to arithmetic progressions of squares, rational right triangles, congruent numbers, and elliptic curves. Among the best-known classical near-misses over $\mathbb{Z}$ are a square of Sallows whose nine entries are distinct squares but for which one diagonal has the wrong sum [21], and fully magic squares of Bremner with all but two entries square [2, 3]. On the structural side, Bruin, Thomas, and Várilly-Alvarado proved that the surface parametrizing 3×3 magic squares of squares is algebraically

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quasi-hyperbolic [4]. Rome and Yamagishi proved that $n \times n$ magic squares of squares exist for every $n \geq 4$ [19]; thus order 3 remains the exceptional unresolved case.

This note studies the order-3 problem over number fields rather than over $\mathbb{Z}$, in the *center-zero* sector, where the center entry is 0. Over a field K , a *magic square of squares* is a 3×3 magic square whose nine entries are squares of elements of K ; it is *proper* if the nine entries are pairwise distinct. We give both an obstruction and a construction. The obstruction: no quadratic number field admits a proper center-zero magic square of squares (Theorem A). The construction: for every congruent number n — recall that a positive integer is *congruent* if it is the area of a right triangle with three rational sides — the biquadratic field $\mathbb{Q}(i, \sqrt{n})$ admits infinitely many proper center-zero magic squares of squares, pairwise inequivalent under scaling and the symmetries of the grid, and degree 4 is the least degree of a number field admitting such a square (Theorem B). For $n = 5$ the construction is completely explicit; see (2) in the sequel.

The question also makes sense over other rings, and the center-zero sector is prominent there. Cain [5] showed that $\mathbb{F}_{29}$ is the smallest finite field admitting a proper magic square of squares and described the center-zero solutions in terms of three consecutive squares [5, Lemma 5.4 and Theorem 5.1]; his minimal example over $\mathbb{F}_{29}$ is itself center-zero; see also Labruna [13]. Wolird [25] studied the Gaussian-integer formulation and transported known integer near-misses to $\mathbb{Z}[i]$. In a parallel direction, arithmetic progressions of squares over quadratic and biquadratic number fields have been studied by Xarles [26], González-Jiménez and Xarles [9], and González-Jiménez and Tho [8]; their works concern longer progressions, whereas the construction behind Theorem B uses two matched three-term progressions.

To state our results precisely, we first sketch the reduction that gives both theorems their shape; Section 2 carries out the argument in detail. By a classical observation of Lucas (Lemma 2.1), a 3×3 magic square over a field K is determined by its center entry e and two further parameters a and b , and when $e = 0$ its nine entries are

$$0, \quad \pm a, \quad \pm b, \quad \pm(a+b), \quad \pm(a-b).$$

If all nine are squares in K and the square is proper, then a and $-a$ are nonzero squares, so K contains $i = \sqrt{-1}$; among quadratic number fields this leaves only $\mathbb{Q}(i)$. Rescaling by a^{-1} normalizes $a = 1$. Meanwhile, $b = t$ is then itself a nonzero square, say $t = u^2$. The remaining requirements — that $\pm(1+t)$ and $\pm(1-t)$ be squares — say exactly that the projective curve

$$(1) \quad C: \quad \begin{cases} y^2 = u^2 + w^2, \\ z^2 = u^2 - w^2 \end{cases} \quad \subset \mathbb{P}_{[u:y:z:w]}^3$$

has a K -point with $uw \neq 0$ and $(u/w)^2 = t$. Finally, the nine entries are distinct precisely when, in addition, $t \notin \{\pm 1, \pm 2, \pm \frac{1}{2}\}$. We call the points of C that fail these requirements — those with $uw = 0$, or with $(u/w)^2$ among the excluded values — *degenerate*. Proposition 2.3 records the resulting dictionary: proper center-zero magic squares of squares over K , up to scaling by nonzero squares, correspond to the values $t = (u/w)^2$ at nondegenerate points of $C(K)$.

Our obstruction result for quadratic fields is then as follows.

Theorem A. *No quadratic number field admits a proper center-zero 3×3 magic square of squares. In fact, the curve C of (1) has exactly eight $\mathbb{Q}(i)$ -points,*

$$C(\mathbb{Q}(i)) = \{[1 : \epsilon : \delta : 0], [0 : \epsilon : \delta i : 1] : \epsilon, \delta \in \{\pm 1\}\},$$

and all eight are degenerate.

The Diophantine content of Theorem A is classical. The pair of equations (1) is the case $(M, N) = (1, -1)$ of Euler's concordant forms, or the classical *congruum* problem; see Ono [17]. Multiplying the two equations gives $(yz)^2 = u^4 - w^4$, which connects Theorem A to Hilbert's theorem that $x^4 - y^4 = z^2$ has only trivial solutions in Gaussian integers. Najman [16] gave a modern proof of Hilbert's theorem using the elliptic curve $E: y^2 = x^3 - x$ over $\mathbb{Q}(i)$, and indeed the Jacobian of C is $\mathbb{Q}$ -isogenous to E (Lemma 3.1). The curve E is the congruent-number curve for $n = 1$: Fermat's right-triangle theorem, that 1 is not a congruent number, amounts to the assertion that $\text{rank } E(\mathbb{Q}) = 0$. The same method that proves Theorem A determines $E(\mathbb{Q}(i))$ completely and shows that 1 is not a congruent number over $\mathbb{Q}(i)$ either, in the nondegenerate sense appropriate to a field containing i (Corollaries 4.1 and 4.2).

The converse direction turns Fermat's obstruction into a construction.

Theorem B. *Let n be a congruent number, which we may take to be a squarefree positive integer, and consider $K = \mathbb{Q}(i, \sqrt{n})$, which is a biquadratic field. Then writing $E: y^2 = x^3 - x$ and $E_n: y^2 = x^3 - n^2x$, we have*

$$\text{rank } E(K) = 2 \text{ rank } E_n(\mathbb{Q}) \geq 2;$$

and K admits infinitely many equivalence classes of proper center-zero magic squares of squares (where equivalence is taken according to multiplication by nonzero squares and symmetries of the grid). Moreover, degree 4 is minimal: every number field admitting such a square contains $\mathbb{Q}(i)$ properly and therefore has even degree at least 4.

For $n = 5$, the smallest congruent number, the construction produces the square

$$(2) \quad \begin{pmatrix} 41^2 & (49i)^2 & (12\sqrt{5})^2 \\ (31i)^2 & 0^2 & 31^2 \\ (12i\sqrt{5})^2 & 49^2 & (41i)^2 \end{pmatrix} = \begin{pmatrix} 1681 & -2401 & 720 \\ -961 & 0 & 961 \\ -720 & 2401 & -1681 \end{pmatrix}$$

over $\mathbb{Q}(i, \sqrt{5})$, with all eight line sums equal to 0 and nine distinct entries (Section 6). Remark 6.1 records a complementary observation: Bremner's square with eight square entries over $\mathbb{Q}(\sqrt{3})$ [3] becomes a proper, non-center-zero magic square of squares over $\mathbb{Q}(\sqrt{3}, \sqrt{133})$. The content of Theorem B is the systematic congruent-number construction and the sharp degree bound in the center-zero sector. That said, we do not address the original problem over $\mathbb{Z}$ or the general unrestricted-center problem over $\mathbb{Z}[i]$.

The paper is organized as follows. Section 2 proves the Lucas parametrization and the dictionary between center-zero squares and points of C (Proposition 2.3). Section 3 identifies the Jacobian of C up to isogeny. Section 4 proves Theorem A and its congruent-number corollaries, and Section 5 proves Theorem B. Section 6 works out the example (2) in detail, and Section 7 records finite-field counterparts, including the relation to Cain's example over $\mathbb{F}_{29}$.

Except in Lemma 2.1, throughout the paper a *field* has characteristic different from 2. All number fields lie in a fixed algebraic closure of $\mathbb{Q}$, and

$$\text{Sq}(K) := \{s^2 : s \in K\}$$

includes 0. For $d \in \mathbb{Q}^*$ we write $E^{(d)}$ for the quadratic twist of E by d ; for $E: y^2 = x^3 - x$ we use the model $E^{(d)}: y^2 = x^3 - d^2x$.

2. CENTER-ZERO SQUARES AND THE PARAMETER CURVE

This section carries out in detail the reduction sketched in the Introduction. We begin with the classical parametrization of 3×3 magic squares, which goes back to Lucas [15]; see Boyer [1] for the history and [18, 21] for modern uses, as well as [3] over $\mathbb{Q}$ and [5] over finite fields. We record it over an arbitrary field, as the usual derivation is characteristic-free.

Lemma 2.1 (Lucas form). *Over any field K (of any characteristic), every 3×3 magic square has the form*

$$(3) \quad \begin{pmatrix} e+a & e-a-b & e+b \\ e-a+b & e & e+a-b \\ e-b & e+a+b & e-a \end{pmatrix} \quad (a, b, e \in K),$$

with magic sum $3e$; conversely, (3) is magic for all $a, b, e \in K$.

Proof. Consider a magic square with entries $A, \dots, I$, read row by row, with common line sum S , i.e.:

$$(4) \quad \begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}$$

with $S = A + B + C = D + E + F = \dots$. The four lines through the center (middle row, middle column, and the two diagonals) together use every entry once and the center E three additional times, so $4S = 3S + 3E$ and $S = 3E$; no division is needed. Set $e = E$, $a = A - e$, $b = C - e$. The diagonals give $I = e - a$ and $G = e - b$; the first row gives $B = e - a - b$; the outer columns give $D = e - a + b$ and $F = e + a - b$; and the bottom row gives $H = e + a + b$. The middle row and middle column sums are then automatic, and one checks directly that (3) is magic with sum $3e$ for all a, b, e . $\square$

Definition 2.2. A *magic square of squares* over K is a square (3) all of whose entries lie in $\text{Sq}(K)$. It is *proper* if its nine entries are all distinct, and *center-zero* if $e = 0$.

Lemma 2.1 implies that any proper center-zero magic square of squares has entries of the form

$$(5) \quad 0, \quad \pm a, \quad \pm b, \quad \pm(a+b), \quad \pm(a-b).$$

Two immediate constraints follow. In characteristic 2 we have $x = -x$, so (5) contains at most four distinct values and no proper center-zero square exists. Now suppose $\text{char } K \neq 2$. Properness gives $a \neq 0$; if $a = r^2$ and $-a = s^2$, then $(s/r)^2 = -1$. Hence:

a field admitting a proper center-zero magic square of squares contains $i = \sqrt{-1}$.

In particular, among quadratic fields over $\mathbb{Q}$ only $\mathbb{Q}(i)$ is in question, and among finite fields of odd order only $\mathbb{F}_q$ with $q \equiv 1 \pmod{4}$ are possible.

Assume now that $i \in K$. Since $a \in \text{Sq}(K) \setminus \{0\}$, multiplying every entry by $a^{-1} \in \text{Sq}(K)$ normalizes $a = 1$. The quotient $t = b/a$ is again a nonzero square, say $t = u^2$ with $u \in K^*$. Since $i \in K$, the remaining conditions that $\pm(1+t)$ and $\pm(1-t)$ be squares amount to

$$(6) \quad 1 + u^2 \in \text{Sq}(K) \quad \text{and} \quad u^2 - 1 \in \text{Sq}(K),$$

that is, to a K -point of the curve C of (1) with $uw \neq 0$, via $t = (u/w)^2$ on the chart $w = 1$.

It remains to decide when

$$(7) \quad 0, \quad \pm 1, \quad \pm t, \quad \pm(1+t), \quad \pm(1-t)$$

are pairwise distinct. Equating two expressions in the list (7) gives either a contradiction in odd characteristic or a linear equation for t whose coefficient is ± 1 or ± 2 . Conversely, direct substitution shows that a collision occurs for each of the resulting seven values. Thus the square is proper exactly when

$$(8) \quad t \notin \left\{0, \pm 1, \pm 2, \pm \frac{1}{2}\right\},$$

a set of at most seven elements of K . We call a point of $C(K)$ *degenerate* if $uw = 0$ or if $uw \neq 0$ and $t = (u/w)^2$ lies in the set (8). Thus a point is nondegenerate precisely when it yields, through the parameter $t = (u/w)^2$, a proper normalized center-zero magic square. Summarizing, we have:

Proposition 2.3. *Let K be a field with $\text{char } K \neq 2$ and $i \in K$. The assignment $t = b/a$ induces a bijection between proper center-zero magic squares of squares over K , up to scaling by nonzero squares, and elements*

$$t \in K^* \setminus \left\{\pm 1, \pm 2, \pm \frac{1}{2}\right\} \quad \text{with} \quad t, 1+t, t-1 \in \text{Sq}(K),$$

and every such t is of the form $(u/w)^2$ for a nondegenerate point $[u : y : z : w] \in C(K)$, with $C \subset \mathbb{P}^3_{[u:y:z:w]}$ defined by $y^2 = u^2 + w^2$ and $z^2 = u^2 - w^2$ as in (1). Consequently, proper center-zero magic squares of squares over K exist if and only if $C(K)$ contains a nondegenerate point.

Proof. Given t as in the proposition statement, the square (3) with $(e, a, b) = (0, 1, t)$ is proper and center-zero. Conversely, every proper center-zero square is the product of exactly one normalized square of the stated form and a nonzero element of $\text{Sq}(K)$, since $t = b/a$ is invariant under scaling. If we have $\{t, 1+t, t-1\} \subset \text{Sq}(K)$ with $t \neq 0$, choose u, y, z with $u^2 = t$, $y^2 = 1+t$, and $z^2 = t-1$; then $[u : y : z : 1] \in C(K)$ is nondegenerate. Conversely, a nondegenerate point has $w \neq 0$; after setting $w = 1$, its coordinates witness (6) with $t = u^2$ avoiding (8). $\square$

For a fixed value of t there are at most eight points of $C(K)$ above it, corresponding to the sign choices for u, y , and z . Transposition sends the Lucas parameters (a, b) to $(a, -b)$, while a quarter-turn sends them to $(-b, a)$. Hence the rotations and reflections of the grid act on $t = b/a$ through the transformations $t \mapsto -t$ and $t \mapsto -1/t$, and every orbit is contained in $\{\pm t, \pm 1/t\}$.

3. THE JACOBIAN OF C

Proposition 2.3 reduces Theorem A to the determination of $C(\mathbb{Q}(i))$. The following lemma supplies the elliptic-curve structure that makes this determination possible.

Lemma 3.1. *With $P_0 = [1 : 1 : 1 : 0]$ as origin, the curve C of (1) is an elliptic curve over $\mathbb{Q}$. It has good reduction at every odd prime and is $\mathbb{Q}$ -isogenous to*

$$(9) \quad E: y^2 = x^3 - x.$$

Proof. The four points $[1 : \pm 1 : \pm 1 : 0]$ lie in $C(\mathbb{Q})$. The Jacobian matrix of the two defining quadrics is

$$\begin{pmatrix} 2u & -2y & 0 & 2w \\ 2u & 0 & -2z & -2w \end{pmatrix},$$

and in odd characteristic its rank can drop only if every pairwise product among u, y, z, w vanishes. At most one coordinate would then be nonzero, which is incompatible with the two equations of C . Thus the same projective model is smooth over $\mathbb{Z}[1/2]$, so C has good reduction at every odd prime. A smooth complete intersection of two quadrics in $\mathbb{P}^3$ is of genus one; choosing P_0 as origin makes C an elliptic curve.

On the chart $w = 1$, the map

$$(u, y, z) \mapsto (U, V) = (u, yz)$$

has degree 2 onto the smooth projective model of $D: V^2 = U^4 - 1$. Since $U^4 - 1$ is separable, this model has genus one. A nonconstant morphism between genus-one curves induces an isogeny of their Jacobians. The quartic D is birational to

$$E_+: Y^2 = X^3 + 4X$$

through $X = 2(U^2 + V)$ and $Y = 2XU$: indeed, $V = X/2 - U^2$ turns the quartic equation into $U^2 = X/4 + 1/X$, whence $Y^2 = X^3 + 4X$. The standard 2-isogeny with kernel $\{O, (0, 0)\}$ sends E_+ to

$$E_4: Y^2 = X^3 - 16X,$$

which is $\mathbb{Q}$ -isomorphic to E under $(X, Y) = (4x, 8y)$. The induced map on Jacobians, followed by this 2-isogeny, gives

$$C \simeq \text{Jac}(C) \longrightarrow \text{Jac}(D) \simeq E_+ \longrightarrow E_4 \simeq E,$$

and proves the claim. $\square$

Remark 3.2. The curve E is the congruent-number curve for $n = 1$; it has conductor 32 and complex multiplication by $\mathbb{Z}[i]$. As recalled in the Introduction, the system (1) is the case $(M, N) = (1, -1)$ of Euler's concordant forms — the congruum problem of Fibonacci and Fermat — for which we refer to Ono [17] and to Weil's historical account [24, Chs. I–II]. Concretely, a nondegenerate point of $C(\mathbb{Q}(i))$ would produce, upon clearing denominators in $(yz)^2 = u^4 - w^4$, a nontrivial Gaussian-integer solution of $x^4 - y^4 = z^2$, so the impossibility statement in Theorem A also follows from Hilbert's theorem [16]. The proof below instead runs the elliptic-curve argument directly, which in addition determines $C(\mathbb{Q}(i))$ exactly.

4. PROOF OF THEOREM A

With the dictionary of Proposition 2.3 and the identification of Lemma 3.1 in place, the proof of Theorem A is short.

The eight points listed in the statement of Theorem A lie on C and have $uw = 0$, so they are degenerate. We show that there are no other $\mathbb{Q}(i)$ -points.

First we consider the rank. Fermat's right-triangle theorem, together with the classical congruent-number correspondence, gives $\text{rank } E(\mathbb{Q}) = 0$ [10, Ch. I], [24, Ch. II]. The quadratic-extension rank decomposition [11] gives

$$\text{rank } E(\mathbb{Q}(i)) = \text{rank } E(\mathbb{Q}) + \text{rank } E^{(-1)}(\mathbb{Q}),$$

and our chosen model for $E^{(-1)}$ is again E . Thus $\text{rank } E(\mathbb{Q}(i)) = 0$. Lemma 3.1 and isogeny invariance of rank imply $\text{rank } C(\mathbb{Q}(i)) = 0$, so $C(\mathbb{Q}(i))$ is finite.

We now bound C 's order by reduction at two primes. Both 5 and 13 split in $\mathbb{Q}(i)$, and C has good reduction at both primes by Lemma 3.1. A direct count from (1) gives

$$(10) \quad \#C(\mathbb{F}_5) = \#C(\mathbb{F}_{13}) = 8.$$

Indeed, the four points with $w = 0$ survive in each characteristic. On the chart $w = 1$, the square sets are

$$\{0, 1, 4\} \subset \mathbb{F}_5, \quad \{0, 1, 3, 4, 9, 10, 12\} \subset \mathbb{F}_{13}.$$

Inspection shows that $u^2 + 1$ and $u^2 - 1$ are both squares only when $u = 0$; this gives the four points $[0 : \pm 1 : \pm i : 1]$.

For an elliptic curve with good reduction at a prime above p , reduction is injective on prime-to- p torsion [22, Proposition VII.3.1]. Reduction at 5 shows that the only possible prime divisors of $\#C(\mathbb{Q}(i))$ are 2 and 5, with its 2-primary part of order at most 8. Reduction at 13 similarly leaves only 2 and 13. Hence $\#C(\mathbb{Q}(i))$ is a power of 2 dividing 8. Since the eight listed points are distinct, they exhaust $C(\mathbb{Q}(i))$. This proves the second assertion of the theorem.

For the first assertion, Proposition 2.3 now excludes a proper center-zero square over $\mathbb{Q}(i)$; and any quadratic number field supporting such a square would contain i by Section 2, hence would equal $\mathbb{Q}(i)$.

Running the same argument on E itself determines its full Mordell–Weil group over $\mathbb{Q}(i)$.

Corollary 4.1. $E(\mathbb{Q}(i)) = \{O, (0, 0), (\pm 1, 0), (i, \pm(1 - i)), (-i, \pm(1 + i))\} \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$.

Proof. The same rank argument gives $\text{rank } E(\mathbb{Q}(i)) = 0$, and direct counts give $\#E(\mathbb{F}_5) = \#E(\mathbb{F}_{13}) = 8$. The two-prime reduction argument used in the proof of Theorem A therefore bounds $\#E(\mathbb{Q}(i))$ by 8. The listed eight points lie on E . For $P = (i, 1 - i)$ we have $(1 - i)^2 = -2i = i^3 - i$, and the duplication slope is

$$\lambda = \frac{3i^2 - 1}{2(1 - i)} = -1 - i.$$

Consequently $2P = (0, 0)$, so P has order 4; its conjugate and their negatives give the other three points of order 4. Together with O and the rational 2-torsion, these exhaust the group and give the asserted structure. This computation also appears in Najman's treatment of the Gaussian quartic equations [16]. $\square$

Corollary 4.1 yields the congruent-number statement promised in the Introduction. Recall that for a field K of characteristic $\neq 2$ and $n \in K^*$, one says n is a *congruent number over K* if there is a right triangle with all three side lengths in K^* and area n . By the classical congruent number correspondence (see [10, Ch. I]; the algebra is valid over any such K), this holds if and only if $E_n: y^2 = x^3 - n^2x$ has a K -point with $y \neq 0$ and $x^2 \neq -n^2$, the triangle being

$$(11) \quad (\alpha, \beta, \gamma) = \left(\frac{x^2 - n^2}{y}, \frac{2nx}{y}, \frac{x^2 + n^2}{y} \right), \quad \alpha^2 + \beta^2 = \gamma^2, \quad \frac{1}{2}\alpha\beta = n$$

(note that $y \neq 0$ forces $x \neq 0, \pm n$). Conversely, a nondegenerate triangle gives

$$x = \frac{n(\alpha + \gamma)}{\beta}, \quad y = \frac{2nx}{\beta};$$

these formulas invert (11). Here $\alpha + \gamma \neq 0$, since otherwise the Pythagorean identity would force $\beta = 0$.

Corollary 4.2. *1 is not a congruent number over $\mathbb{Q}(i)$.*

Proof. By Corollary 4.1, the points of $E(\mathbb{Q}(i))$ with $y \neq 0$ have $x = \pm i$, hence $x^2 = -1$. Formula (11) gives $\gamma = 0$, so these points correspond only to degenerate triangles. Thus no nondegenerate right triangle of area 1 has all three sides in $\mathbb{Q}(i)$. $\square$

5. PROOF OF THEOREM B

We prove Theorem B in four steps: a reduction to squarefree n , the rank formula, the construction of the magic squares, and their infinitude; the minimality assertion then combines the results of Section 2 with Theorem A.

We first justify the normalization in the statement. If $N = nr^2$ is congruent, with n squarefree and $r \in \mathbb{Q}^*$, then scaling a triangle shows that n is congruent, $\mathbb{Q}(\sqrt{N}) = \mathbb{Q}(\sqrt{n})$, and

$$(X, Y) \mapsto (r^2X, r^3Y)$$

identifies E_n with E_N over $\mathbb{Q}$. We may therefore assume that n is squarefree. It is not a square, since otherwise scaling a triangle of area n would make 1 congruent over $\mathbb{Q}$, contradicting Corollary 4.2. Thus $n > 1$ and K is biquadratic, with quadratic subfields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{n})$, and $\mathbb{Q}(\sqrt{-n})$.

The rank formula. For a multiquadratic extension, the Mordell–Weil group tensored with $\mathbb{Q}$ decomposes into eigenspaces under the Galois group, and the eigenspaces are identified with the twisted Mordell–Weil groups; for $K/\mathbb{Q}$ biquadratic this gives

$$\text{rank } E(K) = \sum_{d \in \{1, -1, n, -n\}} \text{rank } E^{(d)}(\mathbb{Q}),$$

see Kramer [11] for quadratic extensions and Rubin–Silverberg [20, Lemma 2.1] for the general multiquadratic statement. With our chosen equations, $E^{(1)}$ and $E^{(-1)}$ are both given by $y^2 = x^3 - x$, while $E^{(n)}$ and $E^{(-n)}$ are both given by E_n . These coincident pairs of models nevertheless represent four distinct character eigenspaces in the decomposition. Since $\text{rank } E(\mathbb{Q}) = 0$, we obtain

$$\text{rank } E(K) = 2 \text{ rank } E_n(\mathbb{Q}).$$

For squarefree n , the torsion of $E_n(\mathbb{Q})$ is exactly the rational 2-torsion, so n is a congruent number over $\mathbb{Q}$ if and only if $\text{rank } E_n(\mathbb{Q}) \geq 1$ [10, Ch. I]. Hence $\text{rank } E(K) \geq 2$.

The construction. Choose a right triangle (α, β, γ) with positive rational sides and area n . The squares

$$\left(\frac{\alpha - \beta}{2}\right)^2, \quad \left(\frac{\gamma}{2}\right)^2, \quad \left(\frac{\alpha + \beta}{2}\right)^2$$

form an arithmetic progression with common difference $\frac{1}{2}\alpha\beta = n$; this is the classical congruence correspondence. Scaling by c^2 with $c \in \mathbb{Q}^*$ produces three squares $\rho^2, \sigma^2, \tau^2 \in \mathbb{Q}$ in arithmetic progression with any prescribed common difference of the form $d = nc^2$. Given such a progression, take $(e, a, b) = (0, \sigma^2, d)$ in (3):

$$(12) \quad \begin{pmatrix} \sigma^2 & -\tau^2 & d \\ -\rho^2 & 0 & \rho^2 \\ -d & \tau^2 & -\sigma^2 \end{pmatrix}, \quad \rho^2 = \sigma^2 - d, \quad \tau^2 = \sigma^2 + d.$$

Every entry of (12) is a square in K : this is clear for ρ^2, σ^2, τ^2 ; their negatives are squares because $i \in K$; and $\pm d = \pm nc^2$ are the squares of $c\sqrt{n}$ and $ci\sqrt{n}$. Moreover,

$$t = \frac{d}{\sigma^2} = \frac{2\alpha\beta}{\alpha^2 + \beta^2}.$$

The arithmetic–geometric mean inequality gives $0 < t \leq 1$, with equality only if $\alpha = \beta$. Equality is impossible for a rational right triangle, since it would give $\gamma/\alpha = \sqrt{2}$. Thus the only value from (8) that could occur is $t = \frac{1}{2}$. But then $\tau^2 = (1 + t)\sigma^2 = \frac{3}{2}\sigma^2$, which would make $3/2$ a rational square. Hence (12) is proper.

Infinitude. Since $\text{rank } E_n(\mathbb{Q}) \geq 1$, the group $E_n(\mathbb{Q})$ is infinite. The mutually inverse formulas surrounding (11), together with the finitely many choices of signs and order of the legs, therefore give infinitely many rational right triangles of area n . For a fixed hypotenuse γ , the relations $\alpha\beta = 2n$ and $\alpha^2 + \beta^2 = \gamma^2$ admit only finitely many ordered pairs (α, β) , so infinitely many values of γ occur. Taking $c = 1$ in the construction gives

$$t = \frac{n}{\sigma^2} = \frac{4n}{\gamma^2},$$

and therefore infinitely many values of t . Scaling leaves t unchanged, while a grid-symmetry orbit contains at most the four values $\{\pm t, \pm 1/t\}$. The quotient by scaling and grid symmetries is consequently infinite.

Minimality. Let L be a number field admitting a proper center-zero magic square of squares. By Section 2, $i \in L$, so $\mathbb{Q}(i) \subseteq L$; and $L \neq \mathbb{Q}(i)$ by Theorem A. Hence $[L : \mathbb{Q}] = 2[L : \mathbb{Q}(i)] \geq 4$, an even number, and the fields $K = \mathbb{Q}(i, \sqrt{n})$ attain the bound.

Remark 5.1. Congruent numbers over number fields have been studied extensively. Tada [23] characterized congruence over real quadratic fields in terms of rational congruent-number data, and Gironde, González-Diez, González-Jiménez, Steuding, and Steuding [7] proved that every positive integer becomes congruent over some real quadratic field and over an explicit cubic field. Theorem B is a structured biquadratic instance in which the resulting rational points are converted into magic squares: after tensoring with $\mathbb{Q}$, the two new Mordell–Weil eigenspaces over $\mathbb{Q}(i, \sqrt{n})$ are

copies of $E_n(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Q}$, and any one point of infinite order suffices for the construction. For longer arithmetic progressions of squares over such fields, see [26, 9, 8].

6. AN EXPLICIT EXAMPLE OVER $\mathbb{Q}(i, \sqrt{5})$

We now run the construction of Theorem B for $n = 5$ — the smallest congruent number [10, Ch. I]. The right triangle

$$\left(\frac{3}{2}, \frac{20}{3}, \frac{41}{6}\right), \quad \left(\frac{3}{2}\right)^2 + \left(\frac{20}{3}\right)^2 = \left(\frac{41}{6}\right)^2, \quad \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{20}{3} = 5,$$

has area 5. The associated progression $\left(\frac{\alpha-\beta}{2}\right)^2, \left(\frac{\gamma}{2}\right)^2, \left(\frac{\alpha+\beta}{2}\right)^2$ is $\left(\frac{31}{12}\right)^2, \left(\frac{41}{12}\right)^2, \left(\frac{49}{12}\right)^2$ with common difference 5; scaling by $c = 12$ to clear denominators, the squares

$$31^2 = 961, \quad 41^2 = 1681, \quad 49^2 = 2401$$

are in arithmetic progression with common difference $720 = 5 \cdot 12^2$. Taking $(\rho, \sigma, \tau, d) = (31, 41, 49, 720)$ in (12) gives the magic square (2),

$$M = \begin{pmatrix} 1681 & -2401 & 720 \\ -961 & 0 & 961 \\ -720 & 2401 & -1681 \end{pmatrix},$$

whose eight line-sums all vanish:

$$1681 - 2401 + 720 = -961 + 0 + 961 = -720 + 2401 - 1681 = 0,$$

$$1681 - 961 - 720 = -2401 + 0 + 2401 = 720 + 961 - 1681 = 0,$$

$$1681 + 0 - 1681 = 720 + 0 - 720 = 0.$$

The magic square M is the entrywise square of

$$(13) \quad \begin{pmatrix} 41 & 49i & 12\sqrt{5} \\ 31i & 0 & 31 \\ 12i\sqrt{5} & 49 & 41i \end{pmatrix}$$

in $K = \mathbb{Q}(i, \sqrt{5})$, and its entries are the nine distinct numbers

$$-2401, -1681, -961, -720, 0, 720, 961, 1681, 2401.$$

As M is proper, the invariant $t = 720/1681$ avoids (8). And we may also verify the rank statement of Theorem B: the point $(-4, 6) \in E_5(\mathbb{Q})$ has infinite order, since its y -coordinate is nonzero whereas $E_5(\mathbb{Q})_{\text{tors}} = E_5[2](\mathbb{Q})$. Thus $\text{rank } E_5(\mathbb{Q}) \geq 1$, and Theorem B gives $\text{rank } E(\mathbb{Q}(i, \sqrt{5})) \geq 2$.

Remark 6.1. Theorem B produces an infinite family of degree-4 center-zero examples. While we do not have a general construction for 3×3 magic matrices with *nonzero* center, we note that at least one degree-4 example can be obtained. Bremner exhibited the magic square over $\mathbb{Q}(\sqrt{3})$

$$\begin{pmatrix} (22 + 4\sqrt{3})^2 & (17 - 9\sqrt{3})^2 & (5 + 13\sqrt{3})^2 \\ (23 - \sqrt{3})^2 & 532 & (23 + \sqrt{3})^2 \\ (5 - 13\sqrt{3})^2 & (17 + 9\sqrt{3})^2 & (22 - 4\sqrt{3})^2 \end{pmatrix}$$

with eight square entries [3, p. 289]. Its entries are the nine distinct numbers $532 + k\sqrt{3}$, where

$$k \in \{0, \pm 46, \pm 130, \pm 176, \pm 306\}.$$

Since $532 = (2\sqrt{133})^2$, the same matrix is a proper magic square of squares over the real biquadratic field $\mathbb{Q}(\sqrt{3}, \sqrt{133})$. This gives a complementary degree-4 example

with nonzero center. The sharp lower bound in Theorem B, however, concerns the center-zero case, in which properness forces the ground field to contain i .

More generally, any proper magic square with algebraic entries becomes a magic square of squares after adjoining square roots of its nine entries. The significance of (2) is therefore not existence over an unspecified number field, but its center-zero form, its systematic origin in the congruent-number construction, and its realization over a number field of the smallest possible degree.

7. FINITE-FIELD CHECKS

The parametrization given in Section 2 is uniform in the field, so it can be tested over finite fields, where everything is enumerable. This section records the resulting data and the relation to Cain's example. The proof of Lemma 3.1 uses only equations with coefficients in $\mathbb{Z}[\frac{1}{2}]$, so it applies verbatim over $\mathbb{F}_p$ for every odd prime p ; since isogenous elliptic curves over a finite field have the same number of points, it follows that $\#C(\mathbb{F}_p) = \#E(\mathbb{F}_p)$ for all odd p . Exhaustive enumeration gives the following values, in agreement with the isogeny:

p	5	13	29	37	41	53	61
$\#C(\mathbb{F}_p) = \#E(\mathbb{F}_p)$	8	8	40	40	32	40	72

For $p = 29$ the count $40 = 4 + 4 + 8 \cdot 4$ decomposes as follows: four points at infinity, four affine points with $u = 0$, and eight values $u = \pm 8, \pm 9, \pm 11, \pm 13$ admitting four sign choices of (y, z) each. These eight values give $t = u^2 \in \{5, 6, 23, 24\}$. None lies in the nonzero part $\{1, 2, 14, 15, 27, 28\}$ of the reduction of (8), so all four parameter classes yield proper center-zero squares over $\mathbb{F}_{29}$. The four classes form a single orbit under the grid symmetries, which act through $t \mapsto -t$ and $t \mapsto -1/t$ (note $5 \cdot 6 \equiv 1$ and $-5 \equiv 24 \pmod{29}$). Cain's example

$$\begin{pmatrix} 9^2 & 11^2 & 1^2 \\ 6^2 & 0^2 & 14^2 \\ 12^2 & 16^2 & 8^2 \end{pmatrix} \equiv \begin{pmatrix} 23 & 5 & 1 \\ 7 & 0 & 22 \\ 28 & 24 & 6 \end{pmatrix} \pmod{29}$$

from [5, §5] is center-zero with Lucas parameters $(a, b) = (23, 1)$, hence corresponds to $t = b/a = 1/23 \equiv 24 \pmod{29}$ in our normalization.

Remark 7.1. Cain's Theorem 5.1 — every finite field of order less than 29 admits no proper magic square of squares at all [5] — subsumes the center-zero case over finite fields. Our parametrization gives a quick independent check. As noted in Section 2, center zero in odd characteristic requires $q \equiv 1 \pmod{4}$; properness also requires at least nine distinct squares, hence $(q + 1)/2 \geq 9$. The only candidates below 29 are therefore $q = 17$ and $q = 25$. Over $\mathbb{F}_{17}$ there are four nonzero values of u for which both $u^2 + 1$ and $u^2 - 1$ are squares, and they give $t = u^2 = \pm 1$. Over $\mathbb{F}_{25}$ there are eight such values of u , and every resulting t lies in $\mathbb{F}_5^*$ — these values are all excluded by (8), as in characteristic 5, its nonzero part is precisely $\mathbb{F}_5^*$. Thus $\mathbb{F}_{29}$ is also the smallest finite field admitting a proper center-zero magic square of squares, in agreement with [5]; cf. also [13].

COMPUTATIONAL VERIFICATION

An ancillary Python 3 script, using only the standard library, checks the elementary exact identities and finite-field enumerations used in the text. In particular, the script verifies the point-count table, smoothness of the displayed model at all

$\mathbb{F}_p$ -points for the tabulated primes, the $p = 29$ decomposition and symmetry orbit, the excluded parameters over $\mathbb{F}_{17}$ and $\mathbb{F}_{25}$, both explicit magic squares, the Gaussian duplication calculation, and the right-triangle and arithmetic-progression identities. These computations are supplementary; none replaces a proof in the text.

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